Insurance Stock Selection: Bupa Arabia vs. Tawuniya

Risk-return comparison of Bupa Arabia and Tawuniya, benchmarked against TASI, using Sharpe ratio and CAPM alpha.

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Summary

Objective:

This study aims to compare two stocks in the insurance sector: Bupa Arabia for Cooperative Insurance Company and Tawuniya (The Company for Cooperative Insurance). The preferred option is identified using the Sharpe ratio as the primary criterion and Alpha as a secondary criterion, while performance is benchmarked against the TASI index as a fixed reference. The analysis uses data for the period from October 2021 to October 2026.

Main finding:

Tawuniya outperforms Bupa Arabia on both the Sharpe ratio (0.08 vs. −0.14) and Alpha (12.5% vs. 4.8%).

Source of the difference:

Volatility is similar for both stocks (about 33%), but Tawuniya's return is higher. The gap therefore comes from return rather than from risk.

Benchmark comparison:

Both stocks outperformed TASI, which declined by 4.7% per year over the period.

Conditional recommendation:

Tawuniya is the preferred option within this period and under these assumptions. The results describe past performance and do not guarantee future returns.

Data period:

October 2021 to October 2026, covering daily share prices for Bupa Arabia, Tawuniya, and the TASI index, along with Bupa Arabia's audited consolidated financial statements.

Observation on Bupa Arabia:

The loss ratio rose to 95.8% in 2025, and part of the year's earnings came from the release of reserves. This is worth noting as a risk factor.

Key limitations:

- The analysis covers only five years.
- Prices are not adjusted for dividends, so total returns may be understated for both stocks.
- Results assume a risk-free rate of 6.17%. Using 4% does not change the ranking.

Introduction and Objectives

Background: The Saudi Insurance Sector

The insurance sector is one of the vital pillars of the Saudi national economy. It is undergoing major structural and regulatory transformations aimed at strengthening its stability and increasing its contribution to gross domestic product, in line with Saudi Vision 2030 (Insurance Authority).

Why were Bupa Arabia and Tawuniya selected?

- First, both companies operate in the same sector, insurance.
- Second, both are listed on the Saudi Exchange (Tadawul).
- Third, both publish financial statements, which makes a comparative analysis possible.

Why is TASI used as the benchmark?

TASI is the broad index of the Saudi market. It provides a fixed reference against which the performance of both companies is measured, rather than comparing each stock only with the other.

Why use the Sharpe ratio?

It measures the excess return above the risk-free rate per unit of total risk (volatility). This allows two stocks with different risk levels to be compared on the same basis.

Why use Alpha?

It measures the excess performance relative to what the CAPM model predicts, based on the stock's beta. It complements the Sharpe ratio because it focuses specifically on market-related risk.

Data and Methodology

Data sources:

- Google Sheets (GOOGLEFINANCE): daily closing prices of Bupa Arabia, Tawuniya, and the TASI index.
- Saudi Exchange (Tadawul): Bupa Arabia's consolidated financial statements.

Data Processing

Dates were normalized to the trading day, and the three price series were aligned on common trading days, leaving 1,209 days. Daily log returns were then computed as ln(Pt / Pt−1), yielding 1,208 observations. Monthly returns were derived from month-end closing prices (60 observations). Annual figures were annualized using 252 trading days for daily data and 12 months for monthly data.

Formulas

1. Daily Log Return

$r_t = \ln(P_t / P_{(t-1)})$
where P_t is the closing price on day t, and $P_{(t-1)}$ is the closing price on the previous day.

2. Annual Return

$R = 252 \times \text{mean}(r_t)$
where 252 is the number of trading days per year, and mean(r_t) is the average daily return.

3. Annual Volatility

$\sigma = \sqrt{252} \times \sigma_{daily}$
where σ_{daily} is the standard deviation of daily returns.

4. Beta

$\beta_i = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$
where R_i is the stock's return, R_m is the TASI return, Cov denotes covariance, and Var denotes variance.

5. CAPM

$E(R_i) = R_f + \beta_i \times (R_m - R_f)$
where $E(R_i)$ is the expected return of the stock, R_f is the risk-free rate (6.17%), and R_m is the market return.

6. Sharpe Ratio

$S = (R_i - R_f) / \sigma_i$
where R_i is the annual return of the stock, and σ_i is its annual volatility. A higher ratio indicates greater return per unit of risk.

7. Jensen's Alpha

$\alpha_i = R_i - E(R_i)$
where $E(R_i)$ is the return predicted by CAPM. A positive alpha means the stock outperformed what CAPM predicts given its beta.

Data and Methodology

Assumptions:

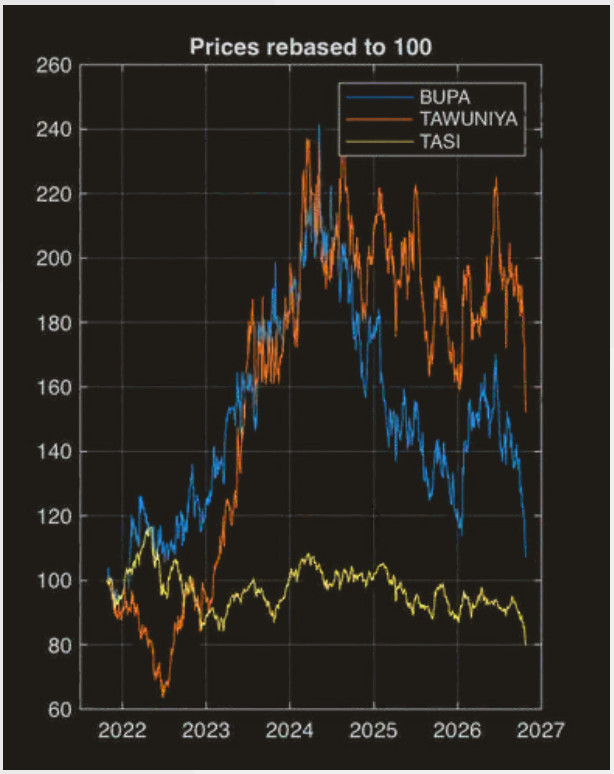
1. The risk-free rate of 6.17% is derived from the yield on government bonds maturing in 2026, and is held constant over the period.
2. The number of trading days per year is 252.
3. Conversion: daily mean × 252 and daily volatility × $\sqrt{252}$ (monthly: × 12 and × $\sqrt{12}$).
4. Returns are computed as log returns.
5. TASI is a price index and does not include dividends.
6. Beta is assumed constant over the period.
7. The simulation assumes that the historical mean and covariance of returns persist.
8. VaR is historical and does not assume a normal distribution.
9. Prices are not adjusted for dividends.
10. The 252-day convention is the standard used for annualization. The aligned sample contains 1,209 common trading days over five years (about 242 per year), because only days on which all three series traded were retained. Using 242 instead of 252 lowers annualized returns by about 4% and volatility by about 2%, which does not change the ranking (Tawuniya's Sharpe ratio stays positive and above Bupa Arabia's).

Tools:

- **MATLAB** with the Statistics and Machine Learning Toolbox and the Financial Toolbox:
  - readtable and readcell: data import
  - synchronize and retime: date alignment and monthly conversion
  - cov and corrcoef: covariance and correlation
  - prctile: historical Value at Risk
  - mvnrnd: multivariate return simulation
- **Microsoft Excel:** data repository, summary tables, and loss ratio calculations.

Market Performance Analysis

Price Trends (Rebased)



Bupa Arabia peaked near 240 in early 2024 before declining to around 107 by October 2026. Tawuniya experienced a deeper drawdown, falling to around 64 in mid-2022, then recovered to about 237 in early 2024 before closing near 152. TASI remained within a range of roughly 80 to 116 and ended near 80, reflecting a broad market decline over the period.

Figure 1: Share prices rebased to 100, October 2021 to October 2026

Descriptive Statistics

	Skewness	Kurtosis
Bupa Arabia	0.21	5.24
Tawuniya	0.24	4.93
TASI	-0.69	8.74

- **Conclusion:**

These results support the use of historical VaR rather than a normal-distribution assumption, and justify highlighting tail risk in the limitations section.

- Bupa Arabia and Tawuniya: Skewness is slightly positive and close to zero (0.21 and 0.24), indicating a nearly symmetric distribution with a marginally longer right tail, meaning large gains occur slightly more often than large losses.
- TASI: Skewness is negative (−0.69), meaning sharp single-day declines are more pronounced than sharp rises.
- Kurtosis: All three series exceed 3, indicating that returns deviate from a normal distribution and extreme movements occur more frequently than expected.
- TASI specifically: The highest kurtosis (8.74), indicating particularly frequent extreme daily movements.

Risk and Return Analysis

- **Expected Returns (Monthly, Annual)**

	Monthly Expected Return	Annual Expected Return
Bupa Arabia	0.09%	1.43%
Tawuniya	0.69%	8.75%
TASI	▼ -0.35%	▼ -4.74%

- **Volatility and Variance**

	Monthly Variance	Annual Volatility
Bupa Arabia	0.0070	33.0%
Tawuniya	0.0090	33.2%
TASI	0.0024	14.0%

Note: Annual volatility is computed from daily returns ($\sigma_{daily} \times \sqrt{252}$), while monthly variance is computed from 60 month-end observations. The two measures use different sampling frequencies, so annualizing the monthly variance does not exactly reproduce the annual volatility shown.

- **Covariance and Correlation**

	Annual Covariance	Correlation
Bupa Arabia-Tawuniya	0.0458	0.42
Bupa Arabia-TASI	0.0172	0.37
Tawuniya-TASI	0.0179	0.38

- **Beta Estimation (vs. TASI)**

	Beta
Bupa Arabia	0.87
Tawuniya	0.91

CAPM and Jensen's Alpha

	Beta	Risk-Free Rate	Market Return (TASI)	CAPM Expected Return	Actual Annual Return	Alpha (Jensen's)
Bupa Arabia	0.87	6.17%	-4.74%	-3.37%	1.43%	4.80%
Tawuniya	0.91	6.17%	-4.74%	-3.76%	8.75%	12.51%
TASI	1.00	6.17%	-4.74%	-4.74%	-4.74%	0.00%

The positive alphas for both companies indicate that their performance exceeded the return predicted by CAPM, given their respective betas, over the study period. The negative expected returns result from the risk-free rate (6.17%) being higher than the market return during the same period. These figures should therefore be interpreted in this context and not as forward-looking expectations.

Stock Selection: Sharpe Ratio and Alpha Comparison

	Annual Return	Voltility	Sharpe Ratio	Alpha (Jensen's)	Rank
Tawuniya	8.75%	33.2%	0.08	12.51%	1
Bupa Arabia	1.43%	33.0%	-0.14	4.80%	2
TASI	-4.74%	14.0%	-0.78	- -	--

Sensitivity check (risk-free rate of 4%):

	Sharpe Ratio
Tawuniya	0.14
Bupa Arabia	-0.08
TASI	-0.62

Conclusion:

Tawuniya ranks first on both the primary criterion (Sharpe ratio) and the secondary criterion (Alpha). The ranking remains unchanged when the risk-free rate is set to 4%, which supports the robustness of the result. Both stocks outperformed the benchmark, which declined over the period. The result describes past performance during October 2021 to October 2026 and does not guarantee future returns.

Downside Risk: Value at Risk and Conditional VaR (Daily, 95%)

	Daily VaR (95%)	CVaR (95%)
Bupa Arabia	-3.13%	-4.49%
Tawuniya	-3.10%	-4.39%
TASI	-1.42%	-2.16%

- Tawuniya shows a slightly lower expected tail loss than Bupa Arabia (CVaR −4.39% vs. −4.49%), while daily VaR is nearly identical, so the difference in risk-adjusted performance is driven mainly by return rather than tail risk.

Actuarial Analysis of Bupa Arabia

Claims Experience and Loss Ratio

	2023	2024	2025
Insurance revenue	15,887.7	18,101.5	19,303.1
Incurred claims	14,737.8	16,674.2	18,487.7
Loss ratio	92.8%	92.1%	95.8%
Expense ratio	2.3%	2.4%	2.3%
Combined ratio	95.1%	94.5%	98.1%

Combined ratio = loss ratio + expense ratio. A combined ratio below 100% indicates underwriting profit. The 2025 figure (98.1%) is close to the break-even level.

Liability for Incurred Claims and Reserve Development

- Liability for incurred claims at year-end 2025: SAR 4,748.6 million, consistent with the figures reported by the external auditor.
- Reserve releases (favourable prior-year development) rose from SAR 552.2 million in 2024 to SAR 1,162.2 million in 2025, equal to 25.5% of the opening reserve.

Ruin Probability (10-Year Horizon)

Claims Volatility(CV)	Probability of ruin
5%	0.0%
10%	3.5%
15%	15.7%

- Ruin probability was estimated by Monte Carlo simulation over a 10-year horizon.
- Assumptions: initial capital equals 2025 equity (SAR 5,673 million), premiums and expense ratio held constant, and no investment income included. The results are therefore conservative.

Limitations

- Study period: The analysis covers just five years and does not capture complete market cycles.
- Prices not adjusted for dividends: Returns are based only on price changes.
- Risk-free rate: The 6.17% rate is taken from the yield of government bonds maturing in 2026 and stays the same throughout the period. A 4% rate was also tested to check that the ranking does not change.
- Actuarial analysis limited to Bupa Arabia: The actuarial analysis was not carried out for Tawuniya, so it is not included in the stock selection.
- Claims triangle: The data available reports claims by financial year rather than accident year, which means a true development triangle cannot be created.
- Investment income: Investment income is not included in the solvency simulation, making the estimated ruin probability conservative.
- Statistical assumptions: Beta is considered constant during the period. VaR is calculated using historical data without assuming normality. Still, the high kurtosis of returns indicates that extreme losses might be underestimated.

Conclusions and Recommendations

Key findings:

- Twuniya stands out as the preferred choice, as it surpasses Bupa Arabia in both the Sharpe ratio (0.08 compared to -0.14) and Alpha (12.5% versus 4.8%).
- Annual volatility is nearly the same for both companies (close to 33%), so the key distinction comes from return rather than risk.
- Both companies performed better than the TASI index, which fell by about 4.74% per year during the period.
- The ranking does not change when the risk-free rate is adjusted to 4%.

Recommendations:

- Preferred option: Tawuniya stands as the preferred choice for this period, given the stated assumptions.
- Bupa Arabia: Investors need to pay attention to the increase in its loss ratio (95.8% in 2025) and the dependence on reserve releases for earnings.
- Future work: Broaden the analysis to include total returns (with dividends), incorporate claims data for Tawuniya to allow a comparable actuarial review, and examine the results over a longer timeframe.

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